

ASafeSight: A PPE Detection System Using Deep Learning for Worksite Safety Compliance

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1. Introduction

In some industries like construction, where the use of Personal Protective Equipment (PPE) is an important issue, workplace safety is a major concern because failure to follow the rules may result in severe injuries or even deaths. Helmets, safety vests, and masks are important tools to ensure minimal risk; although monitoring adherence manually is ineffective, and it can also be humanly interfered with. ASafeSight would be a project created to design an automated PPE detector based on deep learning and computer vision algorithms. Images of construction settings are analyzed by the system and used to determine the presence of necessary safety equipment for workers. The system uses object detection models to give real-time information about PPE compliance

2. Dataset

The project uses the **Construction Site Safety dataset** from Roboflow. The dataset contains approximately 700+ annotated images of construction environments, with bounding box labels for PPE-related classes such as:

- Helmet / Non-Helmet
- Safety Vest
- Gloves
- Shoes
- Person

The dataset is publicly available under the **CC BY 4.0 license**, allowing academic and research use with proper attribution.

The dataset is pre-split into:

- Training set
- Validation set
- Test set

To improve generalization, data augmentation techniques such as scaling and flipping were applied during training.

3. Methodology

3.1 CNN (Computer Vision)

The core of the system is based on YOLOv8 (Ultralytics), a state-of-the-art object detection model. Transfer learning was used with pretrained weights (yolov8n.pt and yolov8s.pt) to improve training efficiency and performance.

Training configuration:

Image size: 640

Epochs: 10

Model variants: YOLOv8n and YOLOv8s

3.2 NLP Component

A simple NLP module was implemented to convert detection outputs into human-readable safety alerts. For example:

“No hard hat detected.”

“No safety vest detected.”

This component helps transform raw detection results into actionable safety messages.

3.3 RL Component

A basic Reinforcement Learning (RL) agent was implemented to simulate adaptive thresholding. The RL module adjusts detection confidence values dynamically, allowing the system to adapt to different environmental conditions. Although simplified, this component demonstrates the integration of adaptive decision-making into the detection pipeline.

3.4 System Pipeline

The system workflow is as follows:

1. The input image is processed
2. YOLO model detects PPE objects
3. RL adjusts detection confidence
4. NLP generates textual alerts

4. Results

The baseline model (YOLOv8n) achieved the following performance:

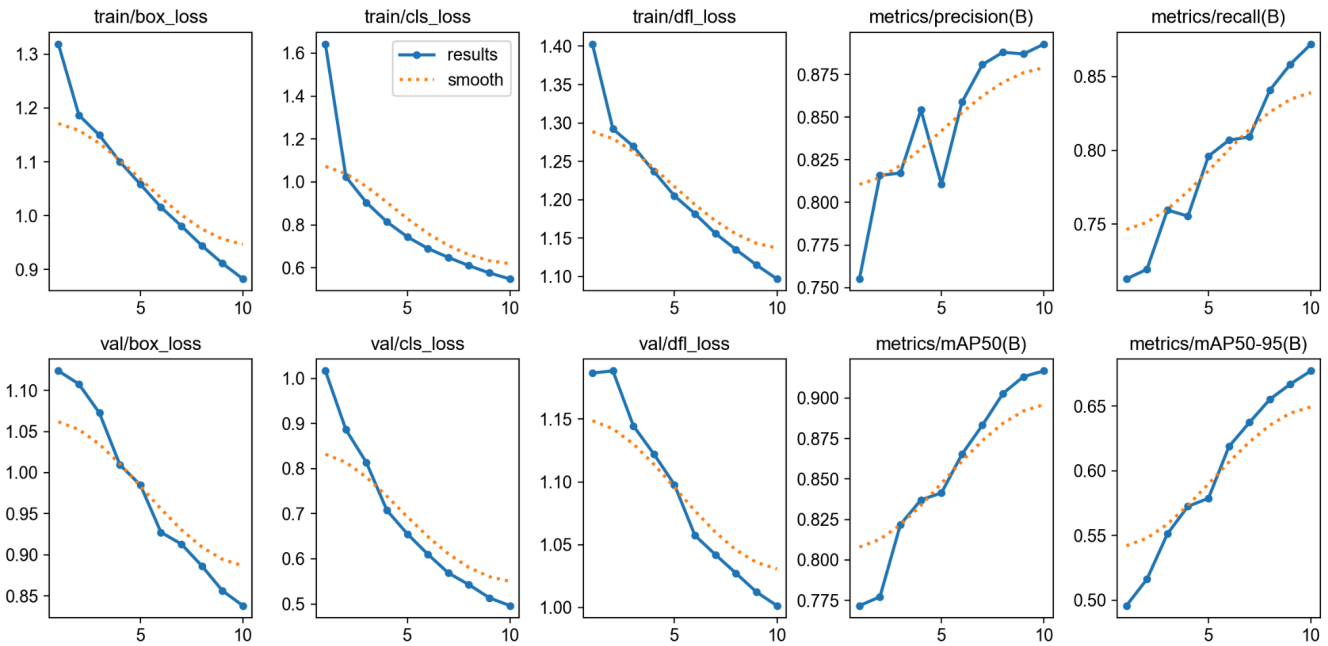
mAP@0.5: 0.92

mAP@0.5–0.95: 0.67

Precision: 0.89

Recall: 0.87

These results indicate strong detection performance and balanced precision-recall behavior.



5. Ablation Study

Two experiments were conducted:

Experiment 1

Model: YOLOv8n

mAP@0.5–0.95: ~0.67

Precision: ~0.89

Recall: ~0.87

Experiment 2

Model: YOLOv8s

mAP@0.5–0.95: 0.736

Precision: 0.88

Recall: 0.95

Analysis

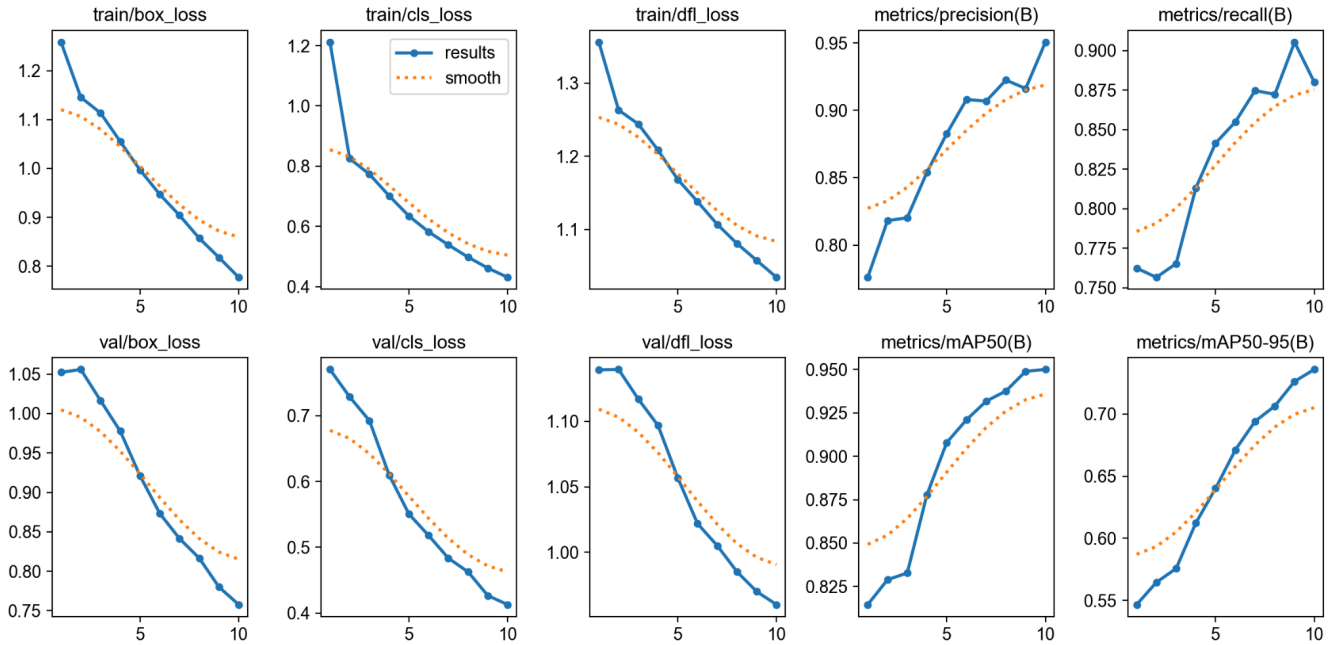
The second model (YOLOv8s) showed improved performance:

Higher mAP indicates better localization accuracy

Recall significantly increased, meaning fewer missed detections

Slight decrease in precision suggests a minor increase in false positives

Overall, the larger model demonstrated superior performance.



6. Error Analysis

Although there was good performance, there were a number of limitations witnessed. The system had difficulties in identifying the personal protective equipment (PPE) in dark conditions, which affected its general performance. There was also a case of missed detections when the objects were partially covered, which decreases the accuracy of the system in such cases. Further, it was less accurate in identifying both small and distant objects, which further restricted its use in some situations. These limitations are primarily due to dataset size and environmental variability.

7. Ethics Considerations

In the system design, there are mitigation measures that are adopted to deal with these ethical issues. First, the system is deliberately created in the manner that preserves privacy, since it does not presuppose any personal identification. It neither does any facial recognition nor keeps any personally identifiable data but only detects PPE devices like helmets, vests, and masks. All processed images are only utilized during real-time inference and are not stored permanently avoiding unnecessary storage, and even in that case, only access by definition.

To minimize the effects of misclassification, the system will offer confidence scores and detection output, which will enable users to determine the accuracy of any particular prediction. This makes sure that the system is involved in the decision-support tool and not the substitute of the human judgment. Also, there are threshold modifications (through the RL component) to enhance sensitivity to detection during changing environmental conditions, to achieve a lower number of false positives and false negatives.

In order to resolve the problem of dataset bias and enhance generalization, data scaling, flipping, and brightness adjustment are used as data augmentation methods during training. These tricks are used to approximate the various conditions of the real world, such as the lighting conditions and the position of the objects. Moreover, the performance of the models is assessed in several classes and conditions to determine the deficiencies, and further enhancement can involve the extension of the sample to cover a wider variety of environments and PPE variations.

Lastly, there is a focus on transparency and responsible deployment. The system is also not aimed at the control of workers or disciplinary measures, but rather aimed at compliance with safety rules. Limitations and proper user guidelines are clearly documented and implemented so that ethical and proper use is achieved in real-world applications.

8. Conclusion

In this project, ASafeSight has managed to prove how deep learning can be used to monitor construction sites and implement automated PPE compliance. The system demonstrated good results by using the YOLOv8 object detection model, high mAP, precision, and recall values, which have shown that the system is reliable in detecting safety equipment and violations. The combination of NLP and RL elements further builds on the system by converting the detection outputs into meaningful safety alerts and the adaptive behavior introduced with the threshold adjustments. The ablation experiment proves that model choice has a vital influence on performance, and the higher-resolution YOLOv8 model has a higher ability to be effective in detecting and recalling objects.

Despite these findings, the system has some constraints associated with the size of the dataset, environments, and detection in adverse conditions, including occlusion and low-light circumstances. On balance, the project shows the possibility of using computer vision along with intelligent components to enhance workplace safety monitoring. Future work is on expanding the data, enhancing the robustness of the model, better adapting the RL mechanism, and developing a more detailed reporting-focused NLP module. ASafeSight can be used as a feasible and scalable aid to the safety compliance in a real-world industry with additional development.

9. References

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